

Revised Final Wastewater Preliminary Engineering Report Addendum

Alturas Wastewater Treatment Plant

Prepared for:

City of Alturas

January 2023

518004.200



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Reference: 518004.200

January 23, 2023

Joe Picotte, Public Works Director
City of Alturas
200 W. North Street
Alturas, CA 96101

**Subject: Revised Final Wastewater Preliminary Engineering Report Addendum
SWRCB Wastewater Planning Grant D17-04002**

Dear Joe Picotte:

Please find enclosed the Final Addendum to the revised final wastewater preliminary engineering report addendum (PER). We have included the revision to Alternative 3, adding nitrogen removal and irrigation, as requested by the Central Valley Regional Water Quality Control Board. This new alternative is called Alternative 3A. We have addressed comments from the State that we received on the draft version of this document. Please review and let us know if you have any further comments. If none, please submit to State Water Resources Control Board Division of Financial Assistance.

If you have any questions, please feel free to contact me at (541) 827-7855 or arasmussen@shn-engr.com.

Respectfully submitted,

SHN

A handwritten signature in blue ink that reads 'Anders Rasmussen'.

Anders H. Rasmussen, PE
Regional Principal/Senior Civil Engineer

AHR:ame

Enclosures: Addendum to PER

c. w/Encl.: Lawrence Sanchez, SWRCB-DFA
Mike Nilsen, RWQCB
Bryan Smith, RWQCB
David Durette, RWQCB

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Alturas Wastewater Treatment Plant

Prepared for:

City of Alturas



Funding for this project has been provided by the California State Water Resources Board under a Proposition 1 Small Community Wastewater Planning Grant, Agreement No. D17-04002.

Prepared by:



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January 2023

QA/QC: MKF____
Reference: 518004.200

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Abbreviations and Acronyms

Units of Measure

Term	Definition
gpd	gallons per day
gpm	gallons per minute
MGD	million gallons per day
mg/L	milligrams per liter
psi	pounds per square inch

Additional Terms

Term	Definition
ADWF	average dry weather flow
BOD	biological oxygen demand
BNR	biological nutrient removal
City	City of Alturas
CWSRF	Clean Water State Revolving Fund
HDPE	high-density polyethylene
MBBR	moving bed bioreactor
PER	preliminary engineering report
PVC	polyvinyl chloride
RWQCB	Central Valley Regional Water Quality Control Board
SWRCB-DFA	State Water Resources Control Board Division of Financial Assistance
WWTP	wastewater treatment plant



1.0 Background

After initial approval of the revised final preliminary engineering report (PER), submitted on June 18, 2021, the Central Valley Regional Water Quality Control Board (RWQCB) expressed concern over potential groundwater impacts from the recommended alternative (Alternative 3: Relocated Wastewater Treatment Plant). A complete site characterization of the parcel where the wastewater treatment plant (WWTP) will be relocated has not been completed but will be completed during the final design phase as part of the report of waste discharge submittal. Given some uncertainties with the site, the RWQCB requested that Alternative 3 be revised to include irrigation as a method of effluent disposal. Since the main concern is nitrate contamination, the RWQCB, the City of Alturas (City), and SHN decided to include a nitrogen removal unit process as well.

In this addendum to the PER, we present the following:

- Design criteria for the irrigation system
- Analysis of various irrigation options with the recommended option
- Design criteria for nitrogen removal
- Discussion of nitrogen removal technologies with a recommended option
- Alternative 3A, which amends Alternative 3 with an irrigation system and nitrogen removal.

2.0 Irrigation System

2.1 Design Criteria

To review the use of an irrigation system to dispose of treated effluent, several issues needed to be considered:

- Will the irrigated area be used for beneficial or non-beneficial use?
- How much land area is available for irrigation, given the site topography?
- Is the available irrigable land sufficient to dispose of the effluent?
- How much storage volume is needed for winter storage?
- How much land area is available for storage, given the site topography?
- What type of irrigation system is to be used?
- What staff requirements are needed for an irrigation system vis-à-vis staff availability?

A key first question to address is whether the irrigation is for beneficial or non-beneficial use. If for beneficial use, such as for hay production, pasture, or other crop production, then Title 22 requirements must be met, including development of a Title 22 engineering report. The City has opted for a non-beneficial use, which is, simply put, watering the native vegetation on site without any further beneficial use.

The property where the proposed relocated WWTP will be located encompasses approximately 278 acres on hilly terrain. No detailed topographic survey has been completed on the site, but based on field observations, it is estimated that approximately 85-90 acres of land is available for irrigation. This includes a 35-acre relatively flat kidney-shaped area along County Road 54. For the analysis discussed herein, a total area of 87 irrigated acres was assumed.



As discussed in the PER, the design flow for the WWTP is 500,000 gallons per day (gpd), with a peak of 1.0 million gallons per day (MGD). Current average dry weather flow (ADWF) is approximately 330,000 gpd, with a current peak daily flow of 900,000 gpd.

Typical net seasonal irrigation depths for a pasture in this area is about 30 to 35 inches per season in an average year depending on crop and precipitation, with the irrigation season typically running from April to October. For the purposes of this analysis, an irrigation season of 7 months (214 days, or 30 weeks) and an average net irrigated depth of 32.5 inches were assumed.

The net volume of water that can be placed on the irrigation area of 87 acres at a depth of 32.5 inches is 76.8 million gallons. The gross volume of water factors in the irrigation efficiency, which varies by irrigation method and other factors, including wind, relative humidity, air temperature, and vegetation cover. Typical sprinkler application efficiencies vary from 58% to 80%. Using a middle value of 70%, the gross volume of water applied is 109.7 million gallons.

The total amount of treated wastewater produced, excluding any losses from evaporation, over a 30-week irrigation period is 69.3 million gallons based on current ADWF of 330,000 gpd, and 105 million gallons based on the design ADWF of 500,000 gallons per day. This indicates that, with the assumed irrigation area, there is enough land to dispose of the current average daily dry weather flow and up to the full design flow during the irrigation season. Additional calculations are provided in Appendix 1.

When looking at winter flow, storage would be required during the non-irrigation period and then disposed of through irrigation during the irrigation season. However, based on the analysis shown in the previous paragraph, there is not sufficient area to allow for summer disposal of stored winter flows through irrigation. Further, the storage volume needed would be approximately 56.7 million gallons, which is the average total flow from November to March based on the 2008–2018 historical flow data. This winter volume would require a pond water surface area of at least 22 acres assuming storage ponds that are eight feet deep, and an additional 49 acres of irrigation area for summer disposal. Detailed calculations are shown in Appendix 1. Due to the hilly topography of the site, it is difficult to locate areas where ponds of significant volume could be placed, and as discussed above, additional suitable irrigation area is limited on the property.

Based on the information presented in the preceding paragraphs, it was concluded that effluent disposal through irrigation can be accomplished only during the irrigation season (April to October), and, during the remainder of the year, the effluent will be disposed of through percolation ponds.

Percolation ponds will be needed for effluent disposal during the winter (that is, non-irrigation) months (November to March) and will remain the same size as described in the PER under Alternative 3. During final design, additional areas might be investigated to provide alternative percolation areas for winter disposal. One of the key concerns of the RWQCB is concentration of percolation, so, in addition to nitrogen removal, use of additional percolation areas will decrease risks to groundwater. This will be determined during full site characterization studies to be performed during final design.

A lined effluent storage basin will be needed to allow for storage of treated effluent, since the effluent flows continually 24/7 while the irrigation system does not. At a minimum, an average day's volume would be stored prior to running the irrigation system. Further, additional storage is needed for other non-irrigating periods during the irrigation season, such as weekends and holidays. A pond with a storage volume of 1.5 million gallons (4.6 acre-feet) would store 4.5 days of flow at current conditions



and three days of flow at the design flow. The amount of storage would provide a reasonable buffer to accommodate holiday weekends (Memorial Day, Labor Day, and occasionally Independence Day) and not require staff to work extra hours to manage the irrigation system during holidays. This storage volume would require 0.75-1.0 acres of area, assuming a pond depth of 8 feet and 2 feet of freeboard. The lined storage pond would be located adjacent to the percolation ponds described in Alternative 3 in the PER. The storage pond would have an emergency overflow to the percolation ponds if the irrigation system were offline for a significant amount of time.

The objective with the irrigation system, given that it is not for a beneficial use, is to apply the effluent over as broad an area as is available to water native vegetation and minimize risks of potential impacts to groundwater by dispersing the treated effluent. Irrigation efficiency is not an important factor; in fact, the more inefficient, the better, for that maximizes the amount of water to be placed over the area available for irrigation.

2.2 Alternatives Analysis

2.2.1 General

Based on the design criteria presented in the previous section and the site constraints, several irrigation options were investigated. The site constraints include irregularly shaped irrigation areas and sloped terrain. Other considerations included labor requirements. The options listed below are discussed in detail in the following sections:

1. Flood Irrigation
2. Wheel Line Irrigation
3. Center Pivot/Linear Irrigation System
4. Sprinkler Irrigation Grid
5. Water Gun Sprinkler, Traveling and/or Stationary

2.2.2 Option 1: Flood Irrigation

Flood irrigation is a simple, cost-effective method where water can flood a field or run overland (if the field is sloped). It is less popular today in agriculture due to its inefficiency. The inefficiency mainly lies in the propensity for deep percolation and surface runoff. These are both undesired outcomes for Alturas due to potential impacts to groundwater and surface waters. Additionally, flood irrigating is labor-intensive and requires skilled workers who can monitor for over-watering. For these reasons, flood irrigation was not pursued.

2.2.3 Option 2: Wheel Line Sprinklers

Wheel line sprinklers are a common irrigation system, with sprinklers mounted on a wheeled pipe system. They are effective at covering large, rectangular areas, moved by either laborers or a motorized system. The lines are typically moved after each watering to a new area. These systems can handle modest constant slopes. However, the varied topography at the site would make it difficult to run a wheel line; therefore, this option was ruled out.

2.2.4 Option 3: Center Pivot/Linear Irrigation

Center pivot and linear irrigation is very common. These systems cover a large area with little to no labor involved, and cost somewhere in the range of \$100,000 depending on size. Pivot systems operate



in an arc and typically go full circle; linears run in a straight line in a rectangular field. One pivot/linear irrigation system can cover up to 150 acres, but they can only effectively irrigate flat, circular, or rectangular areas. Given that the new site has irregularly shaped areas with varied topography available for irrigation, pivot/linear irrigation systems have been eliminated from further consideration.

2.2.5 Option 4: Sprinkler Irrigation Grid

The minimization of labor is a primary concern; therefore, we investigated a fully fixed irrigation system. Many irrigators use a grid network of small sprinklers, close to the ground. Piping is fixed, either located below ground or on the ground. After a consultation with an expert from Nelson Irrigation, we decided that these small sprinklers were not a good option for such a large area, because they are usually for smaller areas, or large areas with a lower application rate than anticipated for this facility.

2.2.6 Option 5: Water Gun Sprinklers, Traveling and/or Stationary

Water guns are large, high-volume sprinklers. These sprinklers can be either 1) stationary attached to tripods or to risers or 2) traveling on movable carts, which are usually attached to hose reels. The hose reels allow the sprinkler to move along a straight alignment as the reel retracts the hose and the sprinkler during the run time.

One of the main manufacturers of water gun sprinklers is Nelson Irrigation (www.nelsonirrigation.com). The Nelson Big Gun sprinklers range in flow from 25 to 1,200 gallons per minute (gpm) with a throw radius of 75 to 310 feet. These sprinklers are a popular choice among irrigators for irrigating large areas with minimum numbers of sprinklers as well as oddly shaped areas such as the proposed property.

Based on a cursory review of available models, the Nelson 200 Series with 1.3-inch taper bore nozzle (27-degree trajectory) appears to be a good option for this application and was used as a basis for estimating project costs. This model disperses 385 gpm (23,100 gallons per hour) over a 200-foot radius circle (2.9 acres) at a pressure of 60 pounds per square inch (psi), which equates to a gross application rate of 0.29 inches per hour and a net application rate of 0.21 inches per hour assuming a 70% irrigation efficiency. The flow rate and throw radius will depend on the provided pressure. With 87 acres area available for irrigation, 30 stationary sprinkler guns would be used. If using traveling guns, a smaller number of sprinklers would be needed, each with a hose reel. The number of traveling sprinklers is estimated to be about 10, assuming a hose reel length of 1,250 feet, but this number may vary based on site constraints such as topography, access, and location of overhead power lines.

Assuming 30 stationary water gun sprinklers operating in a non-overlapping arrangement on a weekly cycle over 5 working days (Monday-Friday), there would be five zones with six sprinklers each. To irrigate the current daily dry weather flow of 330,000 gpd, the sprinklers would operate for 2.4 hours and apply a net irrigation depth of 0.49 inches. To irrigate the design flow of 500,000 gpd, the sprinklers would need to operate for 3.6 hours and apply a net irrigation depth of 0.74 inches (see Appendix 1).

Refinement to the zones and operating run times can be made during final design and after obtaining some operational experience. The key factor is not to overwater either during a single irrigation cycle or for the season.

Weekend and holiday flows can be disposed of by running the sprinklers for additional time and/or in additional zones over the next working days. An average typical maximum net application depth of one inch per week can be used as an initial guide. This translates to a maximum run time of 4.8 hours for



each zone with a net application rate of 0.21 inches per hour to reach the total depth of 1 inch of net irrigation. Actual field conditions may allow for more irrigation, and this can be determined after some initial operating experience.

The irrigation system would be manually started or set to start on a timer and would automatically shut off when the water level in the storage pond reaches a set minimum.

The irrigation system would consist of the following features:

- Lined storage pond with 1.5 million gallons of storage, as described above
- Irrigation pump
- 15,200 linear feet of distribution piping
- Associated valves
- Automatic controllers with float controls in the storage pond
- 30 water gun sprinklers on stationary stands

Based on the anticipated locations of the irrigation areas, approximately 15,200 linear feet of underground irrigation piping is needed from the irrigation pump station to the water gun sprinklers. We have assumed the piping to be 8-inch-diameter high-density polyethylene (HDPE) or polyvinyl chloride (PVC) piping, but smaller diameter piping may be appropriate in some locations, which will be determined during final design.

Traveling water gun sprinklers could replace some of the stationary sprinklers but would require additional operational effort to move the sprinklers after each cycle. Additional analysis may be warranted during final design and after additional site characterization has been performed. For this analysis, stationary sprinklers are recommended.

2.3 Recommended Option

Based on the preceding analysis, Option 5 with stationary water gun sprinklers is recommended for the new irrigation alternative.

3.0 Nitrogen Removal

3.1 Design Criteria

For this project, the objective is to reduce total nitrogen (all species) to at or below 10 milligrams per liter (mg/L), which is the current drinking water standard, prior to percolation and irrigation of the effluent.

3.2 Alternatives Analysis

3.2.1 General

Nitrogen removal in wastewater is a two-step process. First, ammonia is converted to nitrites and then nitrates (since nitrites are unstable) in an aerobic environment with low biological oxygen demand (BOD). This is the nitrification step. Second, nitrates are converted to nitrogen gas in an anoxic environment with a carbon source. This second step is called denitrification. Both steps use specific bacteria to perform the process. This process is called biological nutrient removal (BNR).



Typically, during the tail end of the BOD removal process in an aerated system (lagoon, aeration basins, and so on), full or partial nitrification can occur. Denitrification often requires a separate unit process, which is the case for aerated lagoons.

Chemical processes can be used for nitrification and denitrification, but these are less common and have additional operational costs.

Another less sophisticated process, wood chip bioreactor, is used to remove nitrates from agricultural runoff and is currently being investigated for use with municipal wastewater treatment. This process involves a buried chamber filled with wood chips. Treated effluent is placed in this chamber and, as it slowly flows through, oxygen is depleted creating an anoxic environment where denitrification can occur.

For this project, two options were investigated:

1. Engineered nitrogen removal process
2. Wood chip bioreactor

3.2.2 Option 1: Engineered Nitrogen Removal Process

This option incorporates an engineered nitrification/denitrification unit process. While there are a few different engineered processes for nitrogen removal, the most common units placed after lagoons are moving bed bioreactors (MBBRs). Two such systems are the NitrOx+D system from Triplepoint Water Technologies (www.triplepointwater.com) and the BioPorts MBBR system by Nexom (www.nexom.com).

The MBBR unit would be installed at the outlet of Pond 2. The process includes two chambers. The first chamber is aerated and completes any nitrification that was not completed in the lagoons. The second chamber is anoxic where denitrification occurs. A carbon source is needed in the second chamber and is typically glycol, which is a liquid that is provided by a chemical feed pump.

The manufacturers recommend that part of the effluent from the last lagoon be recycled to the influent location at the first pond. This location where the raw sewage enters the first lagoon is often anoxic and typically with a high carbon source (BOD in this case). Doing this can improve denitrification and reduce the amount of carbon added in the engineered denitrification unit.

3.2.3 Option 2: Wood Chip Bioreactor

As mentioned earlier, wood chip bioreactors are becoming common in treating agricultural runoff, which is often high in nitrates. However, this is still an emerging science for use in municipal wastewater treatment. A recent full-scale study in Bolton, New York, showed that even with an available carbon source in the wood chips, additional carbon from glycol was needed to produce the desired nitrogen removal.

No design guidelines exist for use of a wood-chip bioreactor with municipal wastewater, so we could not provide certainty about the nitrogen removal without pilot studies. Therefore, this option was eliminated from further consideration.

3.3 Recommended Option

The recommended nitrogen removal system is Option 1, Engineered Nitrogen Removal Process.



4.0 Alternative 3A: Aeration Ponds with Nitrogen Removal, Seasonal Irrigation and Seasonal Percolation

4.1 Description

Alternative 3A is the same as Alternative 3 as described in the 2021 PER with the addition of a nitrogen removal process and seasonal irrigation. Based on the analysis of options described in the previous sections, Alternative 3A consists of all of the Alternative 3 elements plus an engineered nitrogen removal system using MBBR technology, a lined effluent storage pond, and a seasonal irrigation system using stationary water gun sprinklers. A clarification and minor modification are needed for the treatment aeration ponds from what was presented in the 2021 PER. First, the discussion for Alternative 3 in the PER did not make clear that the aeration ponds are in series. Second, the order of the ponds has been changed so that the southernmost pond receives the raw influent.

Table 1 summarizes the key components of Alternative 3A.

Table 1. Alternative 3A Component Description

Item	Additional Description
Decommission the existing WWTP ^a	The existing structures will remain.
New Headworks with new screw press screen	Only coarse screening needed.
New Influent Pump Station	To be located adjacent to the new headworks.
New Force main	8-inch diameter pipe to the treatment pond site
New Aeration Ponds	Two ponds with coarse and fine bubble aeration
New Nitrogen Removal System	Nitrification/denitrification reactor process after the aeration ponds
New Evaporation/Percolation Ponds	Two ponds in series able to infiltrate 0.5 MGD ^b , approximately 2-acre bottom area for each pond.
New Effluent Irrigation System	Irrigation system to dispose of water on site during summer months, to include storage pond, irrigation pump, piping, and big gun sprinklers

a. WWTP: wastewater treatment plant

b. MGD: million gallons per day

The existing WWTP will be entirely decommissioned and left in place. The only components that will continue to be used will be the operations building and the perimeter fencing. All other components, including those included in the 2006 improvements funded by Clean Water State Revolving Fund (CWSRF). The WWTP components that were new, upgraded, or replaced under the CWSRF funding in 2006 were the following:

- New:
 - Secondary Clarifier #2
 - Sludge Drying Bed
 - Clarifier Splitter Box
 - Operations Building



- Upgraded:
 - Trickling Filter
 - Chlorine Contact Chamber
 - Filter Effluent Pump Station
 - Drain Water Pump Station
- Replaced:
 - Sewage Grinder

Figure 1 shows the overall layout of this alternative, from the influent pumping station at the existing WWTP to the location of the new WWTP. An enlarged site plan of the new WWTP, including proposed irrigation areas, is shown in Figure 2. Figure 3 shows a process flow diagram of Alternative 3A.

4.2 Costs

Project costs, including final design and construction are estimated to be \$12,847,000. Table 2 presents line-item costs and supporting documentation can be found in Appendix 1.

Table 2. Alternative 3A Opinion of Probable Project Costs ^a

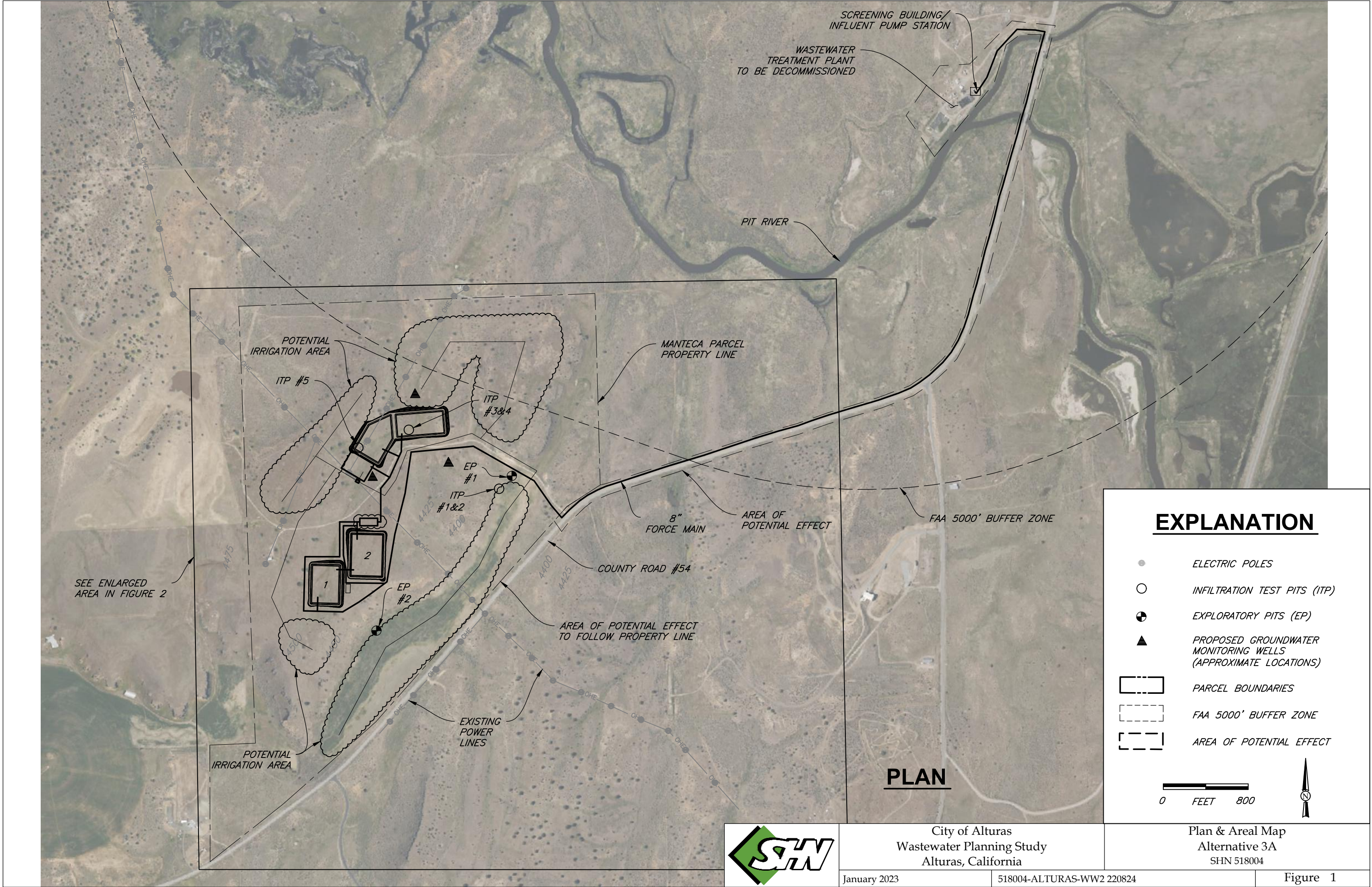
Item	Description	Item Cost
1	Decommission Existing WWTP ^b	\$29,000
2	Backup Generators	\$143,000
3	Flow Monitoring	\$18,000
4	Headworks (Helical Screw Screen)	\$356,000
5	Influent Pump Station	\$1,035,000
6	Influent Force Main	\$670,000
7	Aeration Basins	\$1,193,000
8	Infiltration Ponds	\$180,000
10	Blower Building	\$184,000
11	Nitrogen Removal System	\$1,400,000
12	Lined Irrigation Storage Pond (1.5 million gallons)	\$600,000
13	Irrigation System	\$900,000
	Mobilization (12%)	\$805,000
	Subtotal:	\$7,513,000
	Contingency (30%) ^d	\$2,254,000
	Subtotal Construction:	\$9,767,000
	Land Acquisition	\$361,000
	Groundwater Monitoring Wells ^e	\$80,000
	Engineering (Design) (10%)	\$977,000
	Engineering (Construction Management) (8%)	\$782,000
	Environmental/Permitting (5%)	\$489,000
	Administration/Legal (4%)	\$391,000
	Total Project:	\$12,847,000

a. Items not explicitly listed are incorporated with the listed items or contingency.

b. WWTP: wastewater treatment plant

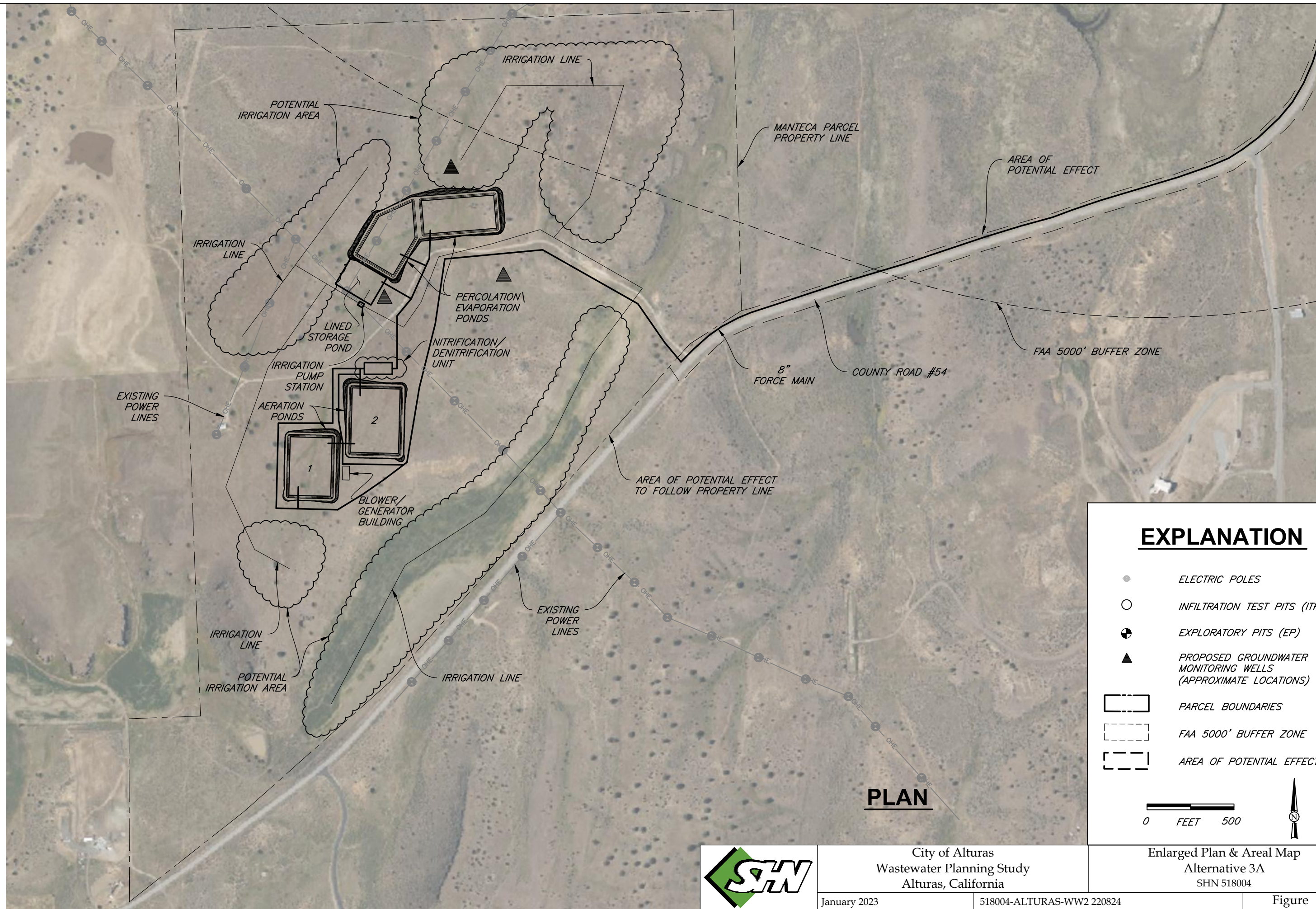
c. Land costs include basic sale at \$1,000/acre plus \$300/acre for closing costs, legal fees, survey, etc.





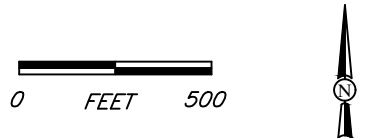
City of Alturas
Wastewater Planning Study
Alturas, California

Plan & Areal Map
Alternative 3A
SHN 518004



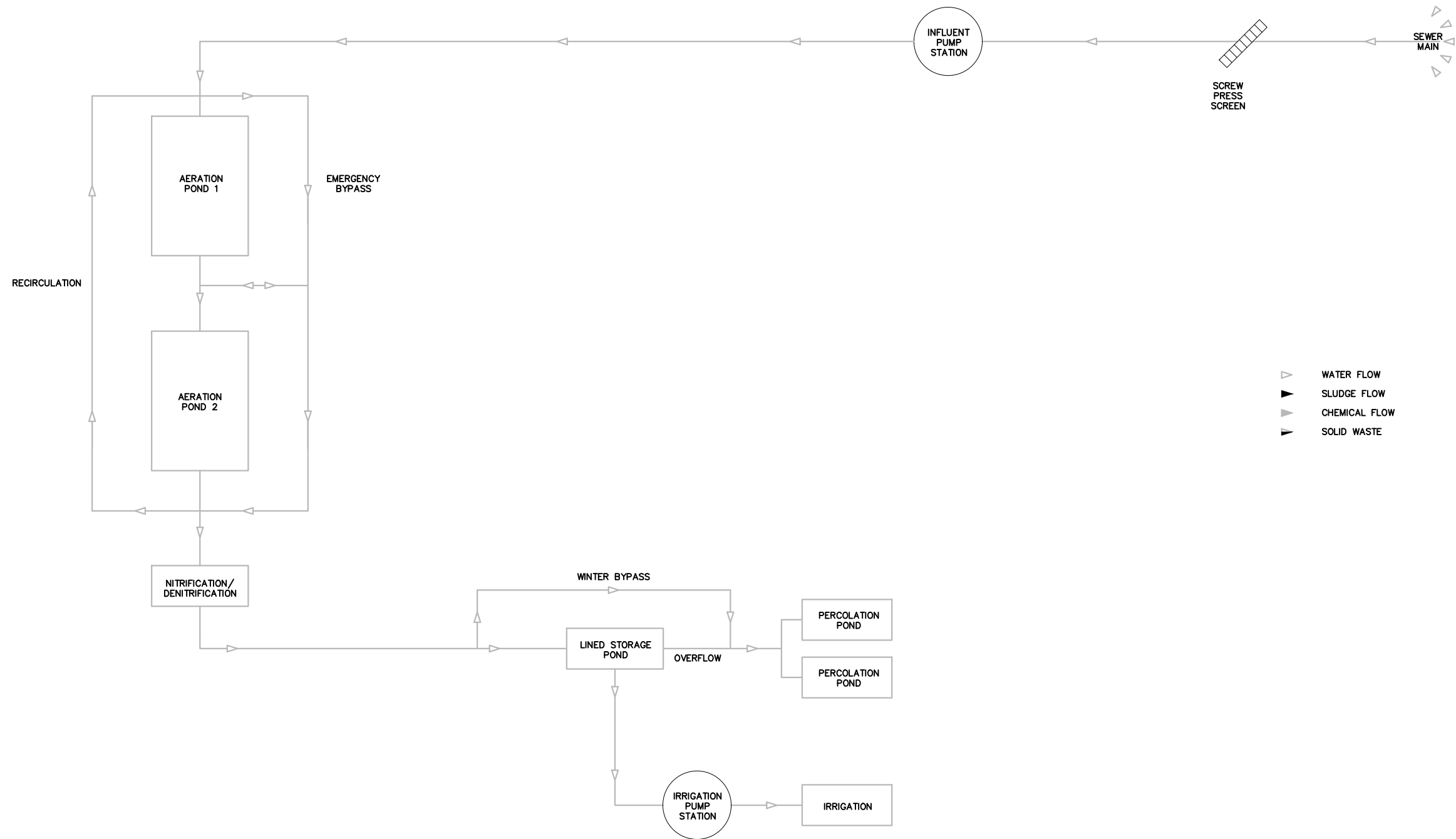
EXPLANATION

- ELECTRIC POLES
- INFILTRATION TEST PITS (ITP)
- EXPLORATORY PITS (EP)
- PROPOSED GROUNDWATER MONITORING WELLS (APPROXIMATE LOCATIONS)
- PARCEL BOUNDARIES
- FAA 5000' BUFFER ZONE
- AREA OF POTENTIAL EFFECT



City of Alturas
Wastewater Planning Study
Alturas, California
January 2023

Enlarged Plan & Areal Map
Alternative 3A
SHN 518004
Figure 2



- d. Contingencies at this planning level are typically at 30% to cover incidental items not specifically covered by the line items as well as unknowns; The grant funding allows for a maximum of 20% contingency to cover for change orders; see Table 3 for additional information.
- e. Costs are for two of the three required monitoring wells. Funding for the first well has been covered through other funding (compliance project).

Funding for the project will likely be through CWSRF. Approximately \$1,265,000 of the funding listed above will be covered by an amendment to the current Wastewater Planning Grant to cover final design, additional site characterization to support the Report of Waste Discharge, additional environmental studies as requested by the State Water Resources Control Board Division of Financial Assistance (SWRCB-DFA), and monitoring wells.

With respect to monitoring wells, the project will require three monitoring wells. Only two are included in the project costs, since the cost of the first monitoring well has been covered by a compliance project approved by the RWQCB. The cost for the first well was approximately \$47,000 and included unanticipated additional costs that are not expected for the other two wells.

The entire project is anticipated to cost approximately \$13,358,000 divided up as follows:

- Planning Grant:
 - Original \$500,000
 - Amendment for design: \$1,265,000
- Construction Grant: \$11,545,000
- Compliance Project: \$48,000

Table 3 shows the construction grant portion in the format required by SWRCB-DFA.

Table 3. Alternative 3A Project Costs for CWSRF Construction Funding Application ^{a,b}

Description	Item Cost
Facilities Construction Cost ^c	\$8,140,000
Pre-Purchase Material Equipment ^d	\$0
Change Order Contingency ^c	\$1,627,000
Construction Management	\$782,000
Value Engineering ^e	\$0
Administration	\$391,000
Land/Right-of-Way	\$361,000
Other Costs ^f	\$244,000
Total Construction Funding	\$11,545,000

- a. CWSRF: Clean Water State Revolving Fund
- b. Costs are from Table 2 and reformatted to match the categories provided by SWRCB-DFA for CWSRF funding
- c. The contingency shown in Table 2 (30%) has been split between Construction and Change Order Contingency so the Change Order Contingency is 20% of Construction; this is reasonable since some of the 30% contingency listed in Table 2 is to cover incidentals that are not listed in the individual line items (such as piping between ponds) and that would normally be part of construction.
- d. All equipment is anticipated to be supplied through the construction contractor and not pre-purchased.
- e. Value Engineering is not anticipated for this project.
- f. Other costs include environmental monitoring and permit compliance and is half of the environmental/permitting item listed in Table 2.



Estimated annual operating costs are \$314,000 and are presented in Table 4. Note that these costs are based on the design flow of 0.5 MGD and are only the core costs for operation of this alternative. The key differences from Alternative 3 in the 2021 PER that cause increased operational costs are 1) chemicals (glycol) needed for nitrogen removal, 2) additional electricity for blowers for the nitrogen removal MBBR system, and 3) electricity for irrigation pumping.

Not included in these operating costs are other administrative costs, loan repayment, and amounts for reserves. A recent rate study prepared by SHN in 2022 use total operating costs of over \$500,000 as part of the basis for the new rates, which have been approved and implemented.

Table 4. Alternative 3A Opinion of Annual Operational Costs ^{a,b}

Description	Item Cost
Labor	\$85,000
Chemicals ^c	\$97,000
Testing/Reporting	\$25,000
Sludge/solids Hauling	\$1,000
Electrical Costs ^c	\$101,000
Permitting	\$5,000
Annual O&M	\$314,000

a. See Appendix 1 for additional detail.

b. Some minor operational costs have not been included because they are assumed to be relatively equal for all alternatives. Detailed operational costs will be shown in a rate study being performed as part of this study.

c. Chemical and electrical costs based on design flow of 0.5 million gallons per day (MGD); anticipated costs for current flow of 0.33 MGD will be less.

5.0 Recommendation

Based on the analysis presented in the 2021 PER and subsequent discussion with the RWQCB, we recommend that Alternative 3A be the preferred alternative.



Irrigation Calculations

1

**CONSULTING ENGINEERS & GEOLOGISTS, INC.**803 Main Street, Suite 401
Klamath Falls, OR 97601

Tel. 541/827-7855

JOB	518004		
SHEET NO.	1	OF	
CALC'ED BY	AHR	DATE	12/29/2022
CHECKED BY	MKF	DATE	1/17/2023

Gross irrigation calculations for new site for Alternative 3A, summer flows only

Estimated Irrigation Area =	87	acres	estimated at 85-90 acres based on field observations
Net seasonal irrigation low =	30	inches	range for pasture is 30-35 inches/season
Net seasonal irrigation mid =	32.5	inches	
Net seasonal irrigation high =	35	inches	
Irrigation Season length =	214	days	April 1 to October 31
	30	weeks	rounded down to nearest number of full weeks

Goal is to irrigate entire area on a weekly basis and rotate through a number of zones.

Assume five-day work week normally, with needed storage over weekends and holidays.

Assume no winter storage, so only flow during irrigation season is disposed of via irrigation.

Wastewater flow data:

current ADWF =	330,000	gallons per day (gpd)
design ADWF =	500,000	gpd

Conversions: 1 acre ft = 325,829 gallons

What is the total net volume that can be placed for the season, based on irrigation need?

Net volume = net irrigation depth x area with conversion

Low net volume = 70,868,000 gallons

Mid net volume = 76,773,000 gallons

High net volume = 82,679,000 gallons

Determine gross volume based on irrigation efficiency, which can range from 58% to 80%.

Gross volume = net volume / irrigation efficiency

Irrigation efficiency = 70% assume middle of range

Low gross volume = 101,240,000

Mid gross volume = 109,676,000

High gross volume = 118,113,000

What total volume of treated effluent is produced during the irrigation season?

Assume the whole number of weeks shown above.

Eff Vol current ADWF = 69,300,000 gallons

Eff Vol design ADWF = 105,000,000 gallons

Total effluent produced is generally less than the gross amount that can be placed via irrigation.

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JOB 518004

SHEET NO. 2 OF

CALC'ED BY AHR DATE 1/19/2023

CHECKED BY DATE

Check requirements for irrigation of annual flow with winter storage

Winter wastewater flow data:

Based on average total volume between November to March from 2008-2018 historical flow data

Total winter volume = 56,700,000 gallons

What pond area is needed to store this amount of water?

Assume ponds with 8 ft water depth.

As initial gross calculation, neglect sloped embankments.

Area = Volume / Depth

Water Depth = 8 ft

Area = 22 acres

What is the irrigation area needed to dispose of the winter volume?

First, how much water can be disposed of per acre?

Net depth of water placed = 30 inches See page 1 for more info

Gross depth of water placed = 43 inches Based on 70% irrigation efficiency

Gross volume per acre = gross depth x 1 acre x 43,560 SF/ac (plus additional conversions)

Gross Vol per Acre = 1,167,553 gallons/acre

Acreage needed to irrigate total winter volume is:

Area needed = 49 acres

Total additional area needed:

Total Area = 71 acres

Does not account for pond embankments.

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JOB	518004	
SHEET NO.	3	OF
CALC'ED BY	AHR	DATE 12/29/2022
CHECKED BY	MKF	DATE 1/17/2023

Detailed typical operational irrigation calculations for new site for Alternative 3A

Based on Nelson Big Gun 200 Series with 1.3-inch taper bore nozzle (27-degree trajectory)

Sprinkler gun data:

Flow =	385	gpm	based on 60 psi pressure being available
Throw radius =	200	ft	
Irrigated area for one gun =	2.9	acres	based on throw radius
Gross application rate =	0.29	inches per hour	
Net application rate =	0.21	inches per hour,	assuming the efficiency from page 1

How many sprinkler guns for the estimated available area?

Estimated Irrigation Area =	87	acres	from page 1
Total guns possible =	30		

On a typical week, there will be five days of irrigation (Mon-Fri).

Divide up the total guns into five zones.

Number of guns per zone =	6
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For the current and design ADWF, how long will it take one zone to apply a day's worth of flow?

Time = ADWF / (gun flow * number of guns per zone)

Time (current ADWF) =	2.4	hours
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Time (design ADWF) =	3.6	hours
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Net irrigation depth applied for above times (based on net application rate show above)

Net Depth (current ADWF) =	0.49	inches
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Net Depth (design ADWF) =	0.74	inches
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Assume Mon-Fri irrigation with weekend and holiday storage (up to three days)

Weekend and holiday stored flows can be metered out the following week evenly, or 2-3 zones per day

can be run. Also, an average typical maximum net application rate of 1 inch/week can be applied to

a given area. This translates to a max run time per zone per week of 1 in / 0.21 in/hr = 4.8 hrs.

This means that a given zone can be run extra time than just for the ADWF.

Storage of 1.5 million gallons, which allow for 3-5 days of storage, depending on the actual flow.

Note, these calculations exclude evaporation effects from the storage basin.

Budgetary Cost Estimates

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Opinion of Probable Project Cost
City of Alturas WWTP
Alternative 3A Cost Summary¹

Item	Description	Units	Quantity	Unit Cost ²	Total Cost
Demolition					
1	Decommission Existing Plant	LS	1	\$ 29,000	\$ 29,000
New Unit Processes					
2	Back-up Generators	LS	1	\$ 143,000	\$ 143,000
3	Flow monitoring	LS	1	\$ 18,000	\$ 18,000
4	Headworks (Spiralift 1/4" Screen w/ Construction and Housing)	EA	1	\$ 356,000	\$ 356,000
5	Influent Pump Station	LS	1	\$ 1,035,000	\$ 1,035,000
6	Influent Force Main	LS	1	\$ 670,000	\$ 670,000
7	Aeration Basin	LS	1	\$ 1,193,000	\$ 1,193,000
8	Infiltration Ponds	LS	1	\$ 180,000	\$ 180,000
9	Blower Building ³	SF	800	\$ 230	\$ 184,000
10	Nitrogen Removal System ⁴	EA	1	\$ 1,400,000	\$ 1,400,000
11	Lined Irrigation Storage Pond (1.5 MG) ⁵	EA	1	\$ 600,000	\$ 600,000
12	Irrigation System ⁶	EA	1	\$ 900,000	\$ 900,000
				Mobilization (12%):	\$ 805,000
				Subtotal :	\$ 7,513,000
				Contingency (30%):	\$ 2,254,000
				Subtotal Construction:	\$ 9,767,000
				Land Acquisition ⁷ :	\$ 361,000
				Groundwater Monitoring Wells ⁸ :	\$ 80,000
				Engineering (Design) (10%):	\$ 977,000
				Engineering (Construction Management) (8%) ⁹ :	\$ 782,000
				Environmental/Permitting (5%):	\$ 489,000
				Administration/Legal (4%):	\$ 391,000
				Total Project:	\$ 12,847,000

Notes:

1. Refer to 2021 PER Alternative 3 costs for details for those items common to Alternative 3.
2. Unless noted otherwise, prices for line items in common with Alternative 3 listed in the 2021 PER have been
3. Blower building has been doubled in size to include blowers for the nitrogen removal system.
4. Nitrogen removal system cost based on doubling of Triplepoint Nitrox+D quote for system components to account for concrete basins, excavation, piping, electrical, controls, and startup.
5. Based on cost of \$0.40/gallon.
6. See attached detail for irrigation system costs
7. Land acquisition costs remain the same as shown in the 2021 PER Alternative 3.
8. Price for two monitoring wells; updated based on recent experience with first well at proposed site.
9. CM to include special inspections.

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JOB	518004		
SHEET NO.	2	OF	
CALC'ED BY	AHR	DATE	10/24/2022
CHECKED BY	MKF	DATE	1/17/2023
		Updated	12/29/2022

Item: Irrigation System

Includes: Irrigation Pump with controls
Distribution Piping
Big Gun sprinklers with risers
Valves, controllers

Notes: 1. Prices shown below based on information obtained from vendors and other sources.
2. Distribution piping assumed to be 8" dia HDPE

Item	Quantity	Unit	Unit Cost	Subtotal	Notes
Irrigation Pump	1	LS	\$ 120,000	\$ 120,000	
Distribution Piping	15,200	LF	\$ 40	\$ 608,000	
Big Gun sprinklers	30	EA	\$ 3,000	\$ 90,000	with risers
Valves, Controller	1	LS	\$ 82,000	\$ 82,000	assumed at 10% of other costs
		Subtotal		\$ 900,000	

For CEQA impacts:

Assume trench 3 ft deep, 1 ft wide. Total excavation is $3 \times 1 \times 15,200 / 27 = 1,689$ CY

Opinion of Annual Wastewater Operational Costs - Alternative 3A
City of Alturas Wastewater Planning Study

Item	Description	Cost
1	Labor	\$ 85,000
2	Chemicals	\$ 97,000
3	Testing/Reporting	\$ 25,000
4	Sludge/solids hauling	\$ 1,000
5	Electrical Costs	\$ 101,000
6	Permitting	\$ 5,000
Annual O&M costs:		\$ 314,000

Notes:

1. Other costs not listed are considered relatively equal between the alternatives.
2. Costs are based on design flow of 0.5 MGD.
3. Values rounded up to nearest \$1,000

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JOB	518004		
SHEET NO.	2	OF	3
CALC'ED BY	AHR	DATE	11/11/2022
CHECKED BY	MKF	DATE	1/17/2023
		updated	1/11/2023

Operations Cost Details for Alternative 3A

Costs from Alternative 3 as shown in the 2021 PER with additions for the irrigation and nitrogen removal systems.

Item No.: 1 Labor

Assume the following for labor including benefits for one full time person.

Per Jason Diven, City of Alturas

Annual Labor: \$ 85,000 per year

- Notes:
1. Labor is only for time as WWTP operator; employee may be covered by other public works items.
 2. Irrigation system is anticipated to operate mostly automatically with little need for staff time.

Item No.: 2 Chemicals

Only chemicals needed is carbon source for denitrification.

Carbon source: MicroC (www.microc.com)

Values based on correspondence with Triplepoint Environmental.

Note per Triplepoint, other less expensive carbon sources may be available, plus recirc will help reduce.

Carbon Source (MicroC):	106 gpd
	\$5.00 per gallon
	\$193,450.00 \$/yr
Total	\$97,000.00 See note below regarding reduction

Per discussion with Tom Daugherty @ Triplepoint, other much less expensive sources they are currently evaluating may be available by the time Alturas comes on line.

Recirculation can also likely reduce carbon need further.

Item No.: 3 Testing/Reporting

Current budget for testing fees is \$35,000.

For Alt 3A, reduced effluent testing but groundwater monitoring; assume 70% of current budget.

Testing: \$ 25,000

Item No.: 4 Sludge/solids hauling

Sludge handling in most recent year was \$973.

Alt 3A includes solids removal at headworks and MBBR unit.

Assume following:

Sludge removal: \$ 1,000

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CALC'ED BY	AHR	DATE	11/11/2022
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Operations Cost Details for Alternative 3A

Costs from Alternative 3 as shown in the 2021 PER with additions for the irrigation and nitrogen removal systems.

Item No.: 5 Electrical Costs

- Notes:
1. Assumed rate of:
\$ 0.17 per kWh Pacific Power
\$ 0.07 per kWh Surprise Valley Electric
Rates cover meter charges for SVE
 2. 1 hp = 0.746 kW
 3. Electrical budget for current FY is \$28,500
 4. Values based on design flow of 0.5 MGD, or 182.5 MG/yr (total flow), 107 MG/yr (irrigation flow)

Headworks plus pumping in Pacific Power service area
Everything else at new WWTP in SVE service area

Headworks screen (allowance):	\$ 600
Influent pumping:	\$ 17,400 560 kWh/MG, 182.5 MG/yr
Pond Aeration	
Blowers (2x50hp=746 kW), running 24/7 =	653,496 kWh per year
Electrical cost:	\$ 45,800 per year
Nitrox+D nite/denite unit (blower and mixer)	
Per Triplepoint Environmental, usage is:	39,016 kWh per month
Electrical cost:	\$ 32,800 per year
Irrigation	\$ 4,200 560 kWh/MG, 107 MG/yr
Assume similar power to effluent pumping	

Total Electrical cost: \$ 101,000**Item No.: 6 Permitting**

Notes: Annual permit estimates were derived from the following fee equation, provided in the California Code of Regulations (Fee Schedule) by the California State Water Resources Control Board:

"Fee equals \$2,572 plus 4,548 multiplied by the permitted flow, in mgd"

$$\begin{aligned}\text{flow} &= 0.5 \text{ mgd} \\ \text{Annual Permit Fee} &= \$ 5,000 *\end{aligned}$$

* The annual permit fee is rounded to the nearest thousands for the planning stage.

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